

Figure Matrix

COMPLETE LEARNING GUIDE

What You'll Learn

Figure matrix questions arrange figures in a 2×2 or 3×3 grid governed by a consistent transformation rule and ask for the single missing figure. This sub-module goes beyond the brief matrix coverage in 'Pattern recognition' to build a complete method: distinguishing row-wise rules from column-wise rules, handling grids where both apply simultaneously, spotting whole-grid (Sudoku-like) constraints, tracking combined transformations (rotation plus element changes together), and using answer-choice elimination as a final check. Each concept is demonstrated with a fully worked grid solved to an exact missing figure.

1. The 2×2 Matrix: Simplest Case

A 2×2 matrix shows 3 figures filling 3 of the 4 cells, with the 4th (usually bottom-right) missing — the rule usually links the two figures in the top row to each other by one transformation, and the same transformation (or a related one) must connect the bottom row, so first identify the top-row transformation and then apply it to the bottom-left figure to derive the answer.

- Top-left is a small circle, top-right is a large circle (the transformation: circle grows larger). Bottom-left is a small square. Applying the same 'grows larger' transformation, the missing bottom-right cell must be a large square.

2. 3×3 Matrix: Row-Wise Rules

In a 3×3 matrix governed by a row rule, each of the 3 rows independently follows the same transformation pattern moving left to right (such as 'number of sides increases by one moving right'), and this pattern must be confirmed using the two complete rows before being applied to find the missing figure in the incomplete third row.

- Row 1: triangle (3 sides), square (4 sides), pentagon (5 sides). Row 2: square (4 sides), pentagon (5 sides), hexagon (6 sides). Both rows confirm 'sides increase by 1 moving right.' Row 3 begins with a pentagon (5 sides) and a hexagon (6 sides); the missing third cell must be a 7-sided heptagon.

3. 3×3 Matrix: Column-Wise Rules

Some matrices instead follow a rule down each COLUMN rather than across each row (e.g., 'shading increases by one section moving downward') — always test both row-wise and column-wise before deciding which axis the rule follows, since applying a column rule's logic across rows (or vice versa) gives a wrong answer even if a pattern superficially seems to fit.

- Column 1 (top to bottom): unshaded circle, half-shaded circle, fully-shaded circle. Column 2: unshaded square, half-shaded square, fully-shaded square. This confirms a column rule: 'shading increases moving down,' independent of row position. Column 3 shows an unshaded triangle and a half-shaded triangle; the missing bottom cell must be a fully-shaded triangle.

4. Matrices With Both Row AND Column Rules

The most demanding matrices apply one rule across rows and a DIFFERENT, independent rule down columns simultaneously — the missing figure must satisfy both rules at once, so solve for what the row rule predicts and what the column rule predicts separately, then confirm they agree on the same final figure.

- Row rule: shape rotates 90° clockwise moving right. Column rule: one small dot is added inside the shape moving down. The missing bottom-right cell must show the bottom-left cell's shape rotated 90° clockwise (row prediction) with one more dot than the middle-right cell (column prediction) — both predictions must describe the same single figure for the answer to be valid.

5. Diagonal and Whole-Grid Rules

Occasionally the rule isn't row- or column-based at all, but instead governs the entire grid as a set — for instance, each of the 3 distinct shapes and each of the 3 distinct shading patterns must appear exactly once in every row and every column (a Sudoku-like constraint) — for these, use elimination: list which shape/shading types are already used in the missing cell's row and column, and the missing figure must be the one type not yet present in either.

- A 3×3 grid uses exactly 3 shapes (circle, square, triangle) and 3 shadings (none, half, full), with each combination appearing once per row and column overall. The missing cell's row already contains a circle and a square (so it must be a triangle), and its column already contains 'none' and 'full' shading (so it must be 'half' shaded). The answer: a half-shaded triangle.

6. Combined Transformations: Rotation Plus Addition

Higher-difficulty matrices combine two transformation types acting on the SAME figure moving in one direction — most often a rotation angle changing together with an element being added or removed — track each transformation as a separate, independent tally (angle in one column, element count in another) rather than trying to see both at once.

- Moving left to right in a row: cell 1 is an arrow pointing up with 1 dot beside it; cell 2 is the same arrow rotated 90° clockwise (now pointing right) with 2 dots; cell 3 continues both patterns — rotated a further 90° clockwise (now pointing down) with 3 dots. The missing figure in the next row, following the identical dual pattern, must show the correct cumulative rotation AND the correct cumulative dot count for its position.

7. Using Answer-Choice Elimination

Once the rule is identified but the exact appearance of the missing figure feels uncertain, use the answer options as a checklist — eliminate any option that violates even one confirmed property of the rule (wrong side count, wrong shading amount, wrong rotation angle), since exam distractor options are deliberately built to satisfy some but not all of the confirmed properties.

- The confirmed rule requires the missing figure to have 5 sides, be half-shaded, and be rotated 45° from the cell above it. Option A has 5 sides and correct rotation but is fully shaded — eliminated. Option B has 5 sides and half-shading but 0° rotation — eliminated. Option C matches all three confirmed properties — selected.

8. Worked Full Example: Solving a 3×3 Matrix End to End

Combine the full method on one grid: scan rows first, scan columns second, check for a whole-grid constraint third, and only after confirming which rule(s) actually hold, derive and verify the missing figure against those rules.

- Row 1: 1 triangle, 2 triangles, 3 triangles (count increases by 1 moving right — confirmed row rule). Row 2: 1 square, 2 squares, 3 squares (same count rule, different shape). Row 3: 1 pentagon, 2 pentagons, and a missing third cell. Applying the confirmed row rule (count increases by 1, same shape as the rest of the row) gives the answer: 3 pentagons.

Common Pitfalls

- Assuming every matrix is row-based without also testing whether the actual rule runs column-wise instead.
- Confirming a row or column rule using only ONE complete row/column instead of at least two before predicting the missing figure.
- In grids with both a row rule and a column rule, satisfying only one of the two rules and ignoring the other.
- Missing a whole-grid Sudoku-style constraint and instead forcing a simpler row or column rule that doesn't actually fit.
- In combined-transformation matrices, tracking rotation and element-count changes together instead of as two separate, independent tallies.
- Picking an answer option that satisfies most but not all of the confirmed rule's properties.
- Not re-verifying the derived answer against BOTH its row and its column before finalizing, in matrices where both apply.

Key Takeaways

- Test both row-wise and column-wise rules before assuming which axis the pattern runs along.

- Confirm any candidate rule against at least two full rows or columns before applying it to the missing cell.
- Some matrices require BOTH a row rule and a column rule to hold simultaneously for the missing figure.
- Watch for whole-grid Sudoku-style constraints where each shape/shading type must appear exactly once per row and column.
- Track combined transformations (rotation + element count) as separate tallies, not as one blended impression.
- Use answer options as an elimination checklist — reject any option that fails even one confirmed rule property.

Learn More

IndiaBix — Non-Verbal Reasoning: Figure Matrix

<https://www.indiabix.com/non-verbal-reasoning/figure-matrix/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>